

TABLE III
Vacuum-type De-soldering Stations from Different Vendors

Specifications	Vendor					
	OK International	GOOT, Japan make	Weller	HAKKO	Srinivasa Electronics	Metcal
Model	MFR-1350	TP-200	WR 2002	Model-474	PLUTO 777	MFR-1351
Temperature controlled (Yes/No)	Not required. Works on better Smartheat technology	Yes	Yes	Yes	Yes	Yes
Power consumption (W)	110	170-200	300 (de-soldering iron 80)	100	200	110
Temperature range (°C)	Not applicable. Works on Smartheat technology	250-450	50-450	380-450	180-480	—
Temperature stability (±°C)	±1.1 (Ideal tip)		±2		1%	±1.1 in still air
Nozzle to ground resistance (Ω)	<2	<2		<2	< 2	<2
Nozzle to ground potential (mV)	<2	<2		<2	0.2	<2
Vacuum generator type	In-built pump		In-built pump with periodic duty 30/30s	Double cylinder type	Diaphragm	Internally
Vacuum pressure (N/m²)	70,000	79,993.2	70,000	80,000	79,993.2	70,000
Heater type and integration (Integrated in the handpiece or separate)	Smartheat-based heater. Never requires tip temperature verification or calibration of the system	Ceramic heater		Ceramic heater -integrated in the handpiece	Separate	Integrated in the handpiece
ESD safe (Yes/No)	Yes	Yes	Yes	Yes	Optional	Yes
Dimensions (mm)	170×200×152.5	95×256×205	273×235×102	165×135×260	—	170×200×152.5
Weight (kg)	—	Total weight of unit: 2.9, De-soldering iron: 0.3	6.7	3.8	4	
Cord length (m)	1.52	1.5	1 (approx.)	1.2	1	1.83

TABLE IV
Hot-Air Rework Systems from Different Vendors

Specifications	Vendor					
	OK International	GOOT, Japan	Weller	HAKKO	Srinivasa Electronics	METCAL
Model	HCT-900	XFC-100		FR-803B	PLUTO 888	HCT-900-21
Temperature controlled (Yes/No)	Yes	Yes	Yes	Yes	Yes	Yes
Power consumption (W)	320	380-420	300, Hot air iron-100	30	200	320
Temperature range (°C)	100-500	150-450	50-550	100-500	180-500	100-500
Airflow (L/min)	5-20	5-20	10	5-20	24	6-25
Dimensions (mm)	210×170×140	227×107×177	273×235×102	160×145×230		210×170×140
Weight (gm)	115	4600	6700 (approx.)	5000	4000	4700
Cord length (m)	1.22	1.1	1 (approx.)	1.2	1	1.83
ESD safe (Yes/No)	Yes	Yes	Yes	Yes	Optional	Yes

name suggests and has a more targeted approach. The parallel movement of its tips allows easy removal of SMD chips and flat packs up to 25mm. Also, the direct heating prevents damage to

neighbouring components on a high-density board.

The third type is called a 'vacuum pump de-soldering station.' This station is mainly used to de-solder through-hole

components, or to remove solder from the pads after the SMD component has been removed using the SMD de-soldering tool. These de-soldering stations are based on the principle of vacuum absorp-